

Choosing and Combining Biological Foundation Models for Sequence Prediction

Literature Review and Methodological Framework

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CAP 6942: Project in Data Analytics, Team Assignment 3

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October 2026

1. Introduction and Organization of the Review

A central problem in computational biology is predicting cellular phenotypes, such as how quickly a messenger RNA (mRNA) is degraded or how strongly a transcript is expressed in different tissues, directly from DNA, RNA, and protein sequences. Earlier work approached this problem through engineered features that summarize known biology, such as codon usage, GC content, RNA folding energy, and regulatory motifs (Presnyak et al. 1111–14; Agarwal and Kelley). In recent years the field has shifted toward biological foundation models: large transformer, convolutional, and state-space networks trained with self-supervised objectives on millions of unlabeled genomic, transcriptomic, or protein sequences (Dalla-Torre et al. 287–89; Lin et al. 1123–26). Each model learns a numerical representation, or embedding, of the sequences it reads, and those embeddings can be reused for prediction tasks the model was never trained on.

Each of these models, however, reads a single molecular alphabet, while phenotype arises from the flow of information between molecules. Chromatin and regulatory DNA control whether a gene is transcribed, splicing and untranslated regions shape the mature mRNA, and translation turns the coding sequence into a protein (Alberts et al. 189–98; Nelson et al., secs. 3.1, 27.1). A recent line of research therefore combines, or fuses, representations from DNA, RNA, and protein models and reports that the combination predicts better than a single model (Mollaysa et al. 1–4; Garau-Luis et al. 1–6; He et al. 942–46).

These early fusion studies leave four questions open, and those questions define our project. First, some studies create the DNA and protein inputs from the same mRNA sequence, so a reported gain may reflect what each model learned during pretraining rather than new biological information (Mollaysa et al. 2). Second, fused models are compared with single models but not with simple sequence statistics such as length and composition, so it is unclear how much of the predictive signal requires a foundation model at all. Third, multimodal machine learning often treats agreement, or alignment, between representations as a goal, yet no fusion study has defined what alignment should mean for DNA, RNA, and protein or measured how much of it reflects shared sequence composition (Huh et al. 20617–19; Tjandrasuwita et al. 59723–25). Fourth, token-level fusion has relied on attention, whose cost grows quadratically with sequence length, and evaluation practices such as single-run results and overlapping training and test sequences make small gains hard to interpret.

This review traces how the field arrived at these questions and where our project takes it next. Section 2 follows the move from biophysical sequence features to single-modality foundation models and identifies what each family of models cannot see. Section 3 compares the main approaches to multimodal fusion. Section 4 examines where that research falls short. Section 5 describes how our project addresses each limitation, from a composition-controlled analysis of frozen encoders to linear-time fusion and a test of whether representation measurements can predict when fusion helps. Section 6 summarizes the gaps and our proposed contribution and states the hypotheses that follow from them.

2. From Biophysical Features to Single-Modality Foundation Models

2.1 Sequence Features That Govern mRNA Stability and Expression

Before deep sequence models, experimental studies established that much of post-transcriptional regulation is written into the sequence itself. Codon optimality, the efficiency with which transfer RNAs decode each codon during translation, emerged as a major determinant of mRNA stability (Presnyak et al. 1111–14). Synonymous codons encode the same amino acid but are decoded at different rates: transcripts rich in optimal codons tend to be stable, whereas non-optimal codons slow elongating ribosomes and are associated with faster mRNA decay (Presnyak et al. 1111–14; Diez et al.). Transcript length, overall GC content, GC content at the third codon position (GC3), RNA folding near the start of translation, and protein-binding motifs in untranslated regions (UTRs) also contribute to stability and translation efficiency (Agarwal and Kelley; Nieuwkoop et al. 2363–67).

Expression also depends on the genome around a gene. Promoters, enhancers, and insulator elements regulate transcription, and chromatin packing controls their accessibility (Alberts et al. 189–98, 334). Splicing removes introns to produce alternative transcript isoforms, and UTRs influence where an mRNA is localized and how long it persists (Alberts et al. 339–40). Models built on hand-engineered features captured individual effects like these, but each feature had to be specified in advance, and such models could not easily represent interactions spread across full transcripts and kilobases of regulatory DNA.

2.2 Single-Modality Foundation Models

Self-supervised pretraining removed the need to specify features by hand and produced three families of models, one for each molecule of the central dogma.

DNA models. The Nucleotide Transformer family (Dalla-Torre et al. 287–91), DNABERT-2 (Zhou et al. 1–6), and HyenaDNA (Nguyen et al. 43177–82) are pretrained on genome assemblies. They differ mainly in how they divide DNA into tokens: Nucleotide Transformer reads six-nucleotide tokens, DNABERT-2 learns variable-length tokens with byte-pair encoding, and HyenaDNA reads single nucleotides with long convolutions that extend its context to one million bases.

RNA and codon models. RNA-FM and RiNALMo are trained with masked language modeling on tens of millions of non-coding RNA sequences (23 million for RNA-FM and 36 million for RiNALMo), mostly from RNACentral, and read one nucleotide per token, which suits RNA secondary structure (Chen et al. 1–3; Penić et al.). CodonBERT (S. Li et al. 1027–30), CaLM (Outeiral and Deane 170–74), and mRNA-FM (ml4bio) are instead trained on protein-coding

sequences read one codon at a time, so their representations can reflect codon usage, translation efficiency, and stability.

Protein models. ESM-2 (Lin et al. 1123–27) and ProtBERT (Elnaggar et al. 7112–18) are trained with masked language modeling on millions of protein sequences from the UniRef databases. ESM-2 spans 8 million to 15 billion parameters, and its representations capture information about protein structure, including contacts between residues, even though the model never sees a structure during training (Lin et al. 1123–26).

2.3 What Each Model Family Cannot See

Each family is limited by the molecule it reads and the way it tokenizes that molecule. Table 1 summarizes the resulting strengths and blind spots.

Table 1. Strengths and blind spots of single-modality foundation model families

Model family	Example models	Tokens	Strengths	Blind spots
DNA	Nucleotide Transformer, DNABERT-2, HyenaDNA	Six nucleotides, byte-pair tokens, or single nucleotides	Promoter and enhancer grammar; regulatory context around a gene when given genomic DNA	Given only a coding sequence, no access to splicing, UTRs, or the surrounding genome
RNA and codon	RNA-FM, RiNALMo, CodonBERT, CaLM, mRNA-FM	Single nucleotides or codons	Synonymous codon usage and translation efficiency (codon models); RNA secondary structure (nucleotide models)	No chromatin or distal regulatory context; no explicit protein structure
Protein	ESM-2 (8M to 15B), ProtBERT	Single amino acids	Structural propensities, residue contacts, and conserved functional sites	No synonymous codon information, because translation maps every synonymous codon to the same amino acid

The table suggests that no single family sees all the information relevant to phenotypes such as stability and expression. A protein model reading a translated sequence cannot distinguish a stable mRNA from an unstable synonymous version that encodes the same protein (Nieuwkoop et al. 2363–66). A nucleotide model, in turn, is not trained to capture the residue contacts that determine how the encoded protein folds (Lin et al. 1123–25). This division of information is the main motivation for multimodal fusion.

3. Multimodal Fusion of Biological Foundation Models

Recent studies combine DNA, RNA, and protein models in three main ways. They differ less in architecture than in what they treat as a modality: a different encoding of the same sequence, a physically different molecule, or a shared vocabulary inside one model.

3.1 Fusion of Views Derived from One Sequence

BioLangFusion (Mollaysa et al. 1–5) creates three inputs from a single coding mRNA: the sequence written in DNA letters, the RNA sequence, and its translated protein. Three frozen

pretrained encoders read these inputs: Nucleotide Transformer v2 (100M) for DNA, RNA-FM for RNA, and ESM-2 (8M) for protein. Because the encoders produce different numbers of tokens, BioLangFusion places all three on a common codon-level grid, averaging RNA-FM's single-nucleotide embeddings in groups of three and upsampling the six-nucleotide DNA embeddings with a learned transposed convolution (Mollaysa et al. 2). It then compares codon-level concatenation, attention pooling across modalities, and cross-attention. Fusion improved on the single encoders across mRNA stability and protein expression tasks, although on mRNA stability the margin was small: the best fused model reached a Spearman correlation of 0.563, against 0.553 for RNA-FM alone (Mollaysa et al. 4).

3.2 Fusion of Distinct Molecules

IsoFormer (Garau-Luis et al. 1–9) instead gives each encoder a physically different molecule in order to predict transcript isoform expression across 30 human tissues from the Genotype-Tissue Expression (GTEx) project (GTEx Consortium 1318–20). Genomic DNA centered on the transcription start site is read by Nucleotide Transformer v2 (12 kb) or by Enformer (about 196 kb; Avsec et al. 1196–98), the full transcript including its UTRs by Nucleotide Transformer v2, and the protein by ESM-2 (150M). Transcripts longer than 12 kb are cropped from the 5' end to keep the 3' UTR, and stacked cross-attention layers let each modality attend to the others, with the cross-attention term set to zero when a transcript has no protein (Garau-Luis et al. 5–6). The encoders are fine-tuned together with the fusion layers. Adding DNA and protein to the RNA input raised R^2 from 0.36 to 0.43, and replacing the Nucleotide Transformer DNA encoder with Enformer raised it to 0.53 (Garau-Luis et al. 7–8).

3.3 Unified Pretraining and Directional Models

Two further approaches move fusion inside the model. LucaOne (He et al. 942–48) is a single 1.8-billion-parameter transformer with 20 encoder blocks, pretrained jointly on nucleic acid and protein sequences with a shared vocabulary of nucleotide and amino acid letters. Central Dogma Transformer (Ota 1–15) builds the direction of biological information flow into its architecture: gene representations from scGPT attend to enhancer-centered DNA representations from a frozen Enformer, and protein representations from ProteomeLM attend to the RNA representations, in order to predict the effects of CRISPRi enhancer perturbations in K562 cells.

Across these approaches a common pattern emerges. Combining modalities is reported to improve prediction, but the studies define a modality differently, train under different regimes (frozen encoders in BioLangFusion, fine-tuned encoders in IsoFormer, joint pretraining in LucaOne), and compare against different baselines. Their results therefore cannot be ranked against one another, and they leave open why and when fusion helps.

4. Where Previous Research Falls Short

These studies show that combining models can raise benchmark scores. Four limitations, however, make it difficult to tell what the gains mean or whether they would hold under stricter evaluation.

4.1 Derived Views Are Not Independent Measurements

BioLangFusion treats the DNA, RNA, and protein forms of one coding sequence as three modalities (Mollaysa et al. 2). Rewriting RNA letters as DNA letters adds nothing, and

translation is a deterministic function of the coding sequence that discards synonymous codon identity (Nelson et al., sec. 27.1). The protein input therefore contains no information that the RNA input lacks. A protein encoder can still help, but only through what it learned during pretraining on protein sequences, such as evolutionary constraints, rather than through a new observation of the cell. Without a control that combines two encoders of the same modality, it is not possible to tell whether a fusion gain reflects cross-modal information or the generic benefit of combining two differently trained models. IsoFormer's genomic DNA input avoids this problem because it carries promoter sequence absent from the transcript, but its protein input is still a translation of the transcript's coding sequence (Garau-Luis et al. 5–6), and neither study separates the two explanations directly.

4.2 Missing Composition and Length Baselines

The fusion studies compare fused models with single-encoder models, but not with simple sequence statistics such as length, GC content, and codon frequencies, even though these statistics are known to predict stability and expression (Presnyak et al. 1111–14; Agarwal and Kelley). Work on probing in natural language processing has shown that a probe's accuracy is meaningful only when compared with controls that reveal what the probe could learn without the representation (Hewitt and Liang 2733–35). Our Stage 1 analysis of twelve frozen encoders shows why such controls matter for biological sequences. A linear model on 87 composition features (GC, GC3, log length, 64 codon frequencies, and 20 amino acid frequencies) explains 61% to 92% of the embedding variance of the four DNA encoders (92.4% for HyenaDNA and 80.4% for Nucleotide Transformer v2), compared with 30% to 68% for the RNA encoders and 42% to 48% for the protein encoders. On GTEx expression, four transcript-length features reach a cross-validated R^2 of 0.242, higher than any single encoder (0.03 to 0.17) and equal to all twelve encoders combined (0.239). The pattern depends on the task: on mRNA stability, length alone predicts almost nothing (R^2 of 0.003), while the protein and codon-level encoders reach 0.11 to 0.14. A fusion gain therefore needs to be judged against composition and length baselines chosen for the task.

4.3 Alignment Is Assumed Rather Than Defined

Alignment, relating corresponding elements across modalities, is a core problem in multimodal machine learning (Baltrušaitis et al. 423–25). Contrastive methods such as CLIP train paired representations to coincide (Radford et al. 8748–50), similarity measures such as centered kernel alignment (CKA) compare representation spaces (Kornblith et al. 3519–21), and the platonic representation hypothesis proposes that models trained on different data converge toward shared representations (Huh et al. 20617–19). Recent theory qualifies this view: Tjandrasuwita et al. find that the alignment achievable between two modalities is limited by how much unique information each carries, and that alignment tracks downstream performance when modalities are redundant but not when unique information dominates (59723–25). DNA, RNA, and protein carry exactly this kind of unique information, such as synonymous codons absent from the protein and promoter sequence absent from the transcript.

Yet none of the fusion studies reviewed here measures how closely the biological encoders agree or what drives their agreement. Our Stage 1 analysis suggests that much of it is shared composition: removing the 87 composition features cuts linear CKA by 52% to 90% for every encoder pair that includes a DNA encoder. Only a few pairs remain strongly aligned after this control, namely the ESM-2 models with one another, RNA-FM with RiNALMo, and CaLM with

the protein encoders, and per-sequence alignment does not predict prediction error for any pair. Raw similarity between frozen embeddings is therefore a weak target for a fusion model, and enforcing it through an alignment loss could pull representations toward shared composition and suppress information that only one modality carries.

4.4 Scaling and Evaluation Practices

Three further practices limit how far current results can be trusted. First, token-level fusion has relied on attention, whose cost grows quadratically with sequence length (Vaswani et al. 5998–6000), while the pretrained encoders also impose context limits: BioLangFusion truncates RNA sequences longer than 1,000 tokens, and IsoFormer crops transcripts longer than 12 kb (Mollaysa et al. 7; Garau-Luis et al. 6).

Second, results are often reported without their variability. BioLangFusion reports one score per model, so its 0.010 margin on mRNA stability cannot be compared with the variation expected across random seeds or data splits; IsoFormer, by contrast, reports means and standard deviations over five seeds (Garau-Luis et al. 7). BioLangFusion also uses each frozen encoder's output without comparing layers, although the Nucleotide Transformer authors found that probing performance was never highest at the final layer (Dalla-Torre et al. 288; Alain and Bengio 1–2).

Third, splits can leak information between training and test data (Kapoor and Narayanan). IsoFormer separates its training and test sets by gene (Garau-Luis et al. 6). The CodonBERT mRNA stability split that BioLangFusion follows is less strict: in the version of the file our project uses, 5,392 of the 8,283 distinct test sequences also appear in the training split, and identical sequences carry labels that differ by a median standard deviation of 0.49, so a model that sees only the sequence can explain at most about 59% of the label variance. Whether this file matches BioLangFusion's filtered version is unresolved, but overlap and label noise of this size mean that published stability scores mix generalization with memorization and have a ceiling well below perfect prediction.

5. Where We Plan to Take the Research

Our project is organized in four stages that address these limitations in turn. Stage 1 has been carried out on two datasets: the CodonBERT mRNA stability data (981 retained sequences), where all three modalities derive from one coding sequence, and a 2,000-transcript sample of IsoFormer's GTEx data (999 training and 996 test transcripts after filtering), where DNA, transcript, and protein are distinct molecules. Stages 2 to 4 build on its findings.

5.1 Composition-Controlled Analysis of Frozen Encoders (Stage 1)

To separate learned biological signal from sequence statistics, Stage 1 analyzes twelve frozen encoders, four per modality: Nucleotide Transformer 500M and v2 100M, DNABERT-2, and HyenaDNA for DNA; RNA-FM, RiNALMo, mRNA-FM, and CaLM for RNA; and ESM-2 at 8M, 35M, and 150M parameters and ProtBERT for protein. Ridge regression probes measure what each embedding makes linearly accessible, using five folds that keep identical sequences, or transcripts of the same gene, together, plus ten further fold assignments that show how much the partition moves each result. Linear CKA, mutual nearest-neighbor overlap, and canonical correlation analysis with cross-modal retrieval compare the encoders, and a layer-by-layer CKA analysis on a subsample compares every hidden state. Every comparison is repeated after

removing the part of each embedding that the 87 composition features predict linearly. For an embedding matrix X , the composition share is

$$R^2_{\text{comp}} = 1 - \|X^\perp\|^2 / \|X_c\|^2$$

where X_c is X with each column centered, $X^\perp$ is what remains of X_c after regressing it on the composition features by ordinary least squares, and $\|\cdot\|$ is the Frobenius norm. Reporting each result before and after this control shows how much of an encoder's signal, and of its agreement with other encoders, goes beyond composition.

5.2 Defining Alignment (Stage 2)

Stage 2 is a decision rather than a computation: it uses the Stage 1 measurements to define alignment before any fusion model is trained. We distinguish positional alignment, where positions in one input correspond to positions in another because the biology fixes them, from representational alignment, where two encoders organize the same examples in the same way, and we keep both separate from predictive complementarity, where two encoders predict better together than either alone. Two encoders count as aligned only to the extent that their representations of corresponding inputs agree on held-out examples after the composition control, and alignment is reported for every pair beside its fusion result rather than used as a training objective. The first fusion models therefore train on the prediction loss alone. Positional alignment is imposed only where the biology supplies it: on coding sequences every token stream can be placed on one codon grid, while on GTEx the promoter window, transcript, and protein have no position-by-position correspondence. Because CaLM with a protein encoder is the one cross-modal pair whose agreement survives the control, that pair will be trained with and without an explicit alignment loss to test this decision directly.

5.3 Linear-Time Fusion with Coupled Mamba (Stage 3)

Stage 3 trains fusion models on token-level embeddings rather than the pooled vectors of Stage 1. Coupled Mamba (W. Li et al. 59808–12) is the priority architecture for coding sequences. It runs one selective state-space chain per modality and couples the chains by summing the hidden states of all modalities before each state transition:

$$h_t^{(m)} = S_m \sum_i h_{t-1}^{(i)} + B_m x_t^{(m)}$$

where $h_t^{(m)}$ is the hidden state of modality m at step t , $x_t^{(m)}$ is its input token, S_m and B_m are learned parameters, and the sum runs over all modalities. Its cost grows linearly rather than quadratically with sequence length. The coupling requires the modalities to share one step index, which the codon grid supplies: single-nucleotide embeddings are averaged over each codon's three positions, each six-nucleotide token is split into its two codons, byte-pair tokens are assigned to the codons they cover, and codon-level RNA and protein encoders already produce one token per codon. On GTEx, where no shared grid exists, IsoFormer-style cross-attention is the primary architecture, and it also serves as Coupled Mamba's competitor on coding sequences. Concatenating the encoders' embeddings and passing them to a small multilayer perceptron is the reference baseline that every fusion model must beat.

5.4 Benchmarking and Predicting When Fusion Helps (Stage 4)

Stage 4 tests the fusion models against a fixed hierarchy of baselines: the training mean, length and composition features, the best single encoder with the same prediction head, a matched

ensemble of two encoders from the same modality, concatenation, and the published BioLangFusion and IsoFormer results. Evaluation uses deduplicated splits that keep identical sequences or genes on one side of the split, repeated fold assignments, and ablations that remove modalities, add an alignment loss, remove the codon grid, residualize composition, or recode a gene with synonymous codons. The tasks begin with mRNA stability and GTEx expression and extend to further mRNA half-life and translation efficiency datasets.

The central test is whether the Stage 1 measurements predict fusion gain. For each encoder combination and task, we measure the gain of the fused model over the best single encoder and ask whether alignment and complementarity, measured before and after the composition control, predict that gain on held-out tasks. If they do, choosing which models to combine becomes a cheap measurement made before any fusion model is trained. If they do not, the result shows that the geometry of frozen representations does not translate into fusion benefit, which is itself useful evidence for the field.

6. Summary

The literature reviewed here moves from hand-built sequence features, through single-modality foundation models that each see part of the central dogma, to fusion methods that combine them. The fusion studies establish that combining DNA, RNA, and protein models can improve prediction, but not why. Derived views mix new information with pretraining priors, missing composition and length baselines leave open how much signal requires a foundation model, alignment is borrowed from other multimodal domains without a biological definition, and evaluation practices make small gains hard to separate from noise or leakage. Our project addresses these gaps by measuring what frozen encoders represent after a composition control, defining alignment from those measurements, fusing encoders in linear time on a biologically grounded grid, and testing whether measurements taken before training predict which combinations will help. Table 2 summarizes the gaps and our response.

Table 2. Limitations of prior fusion research and our proposed contribution

Dimension	Prior research	Limitation	Our contribution
Input framing	BioLangFusion derives DNA and protein inputs from one mRNA; IsoFormer uses genomic DNA, transcript, and protein.	Derived views mix pretraining priors with new information; no same-modality control.	Analyzes derived and distinct inputs separately; adds a same-modality matched ensemble baseline.
Baselines	Fused models compared with single encoders.	No length or composition baseline, although composition explains 61% to 92% of DNA embedding variance and length matches all twelve encoders on GTEx.	87-feature composition control and length baseline reported with every result.
Alignment	General multimodal learning treats alignment as a goal; biological fusion studies do not measure it.	Raw agreement between encoders is largely shared composition.	Alignment defined as agreement after the composition control, reported per pair rather than imposed; alignment-loss ablation.

Dimension	Prior research	Limitation	Our contribution
Fusion mechanism	Concatenation, attention pooling, and cross-attention.	Quadratic attention cost; encoder context limits lead to truncation.	Linear-time Coupled Mamba on a codon grid; cross-attention for distinct inputs; concatenation baseline.
Layer choice	Frozen encoder outputs (BioLangFusion) or full fine-tuning (IsoFormer).	Layer choice not examined, although the final layer was never best in Nucleotide Transformer probing.	Layer-by-layer CKA across the hidden states of all twelve encoders.
Evaluation	One score per model (BioLangFusion); five seeds and gene-separated splits (IsoFormer).	Small gains cannot be separated from split variation or train and test overlap.	Grouped, deduplicated splits; ten further fold assignments; ablation suite.
When fusion helps	Gains reported after training.	No criterion for predicting which combinations will help.	Tests whether Stage 1 measurements predict fusion gain across tasks.

Research Hypotheses

The gaps above lead to five hypotheses for the fusion experiments.

H1. Cross-modal synergy. Fusion outperforms the best single encoder, a matched ensemble of two encoders from the same modality, and length and composition baselines.

H2. Diagnostics predict gains. Encoder pairs with larger Stage 1 concatenation gains and complementarity that survives the composition control, such as a codon-level RNA encoder with a protein encoder, show larger fusion gains than redundant pairs or pairs whose agreement is mainly compositional.

H3. Value of token-level fusion. Token-level fusion with Coupled Mamba or cross-attention outperforms concatenation of pooled embeddings on tasks governed by position-specific sequence features, such as stability and translation efficiency, but not on tasks dominated by overall sequence counts.

H4. No benefit from an alignment loss. Adding an explicit contrastive or cosine alignment loss to the fusion of CaLM and ESM-2 does not improve prediction over the task loss alone.

H5. Length on GTEx. Transcript-length features explain a substantial share of GTEx expression variance, so a fusion model must exceed the length baseline before its gain can be attributed to learned sequence representations.

Together, these stages turn the question raised by the fusion literature, whether combining biological foundation models helps, into a more useful one: when it helps, why, and whether that can be known before a fusion model is trained.

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